

The bug in



$$\sum_v g(\mu, \nu) * T(i, j, \nu)$$

$\neq H(j, i, \mu)$ WRONG !!!

$= H(i, j, \mu)$ OK 😊

In Physics, we are dealing with objects of different tensor rank, such as:

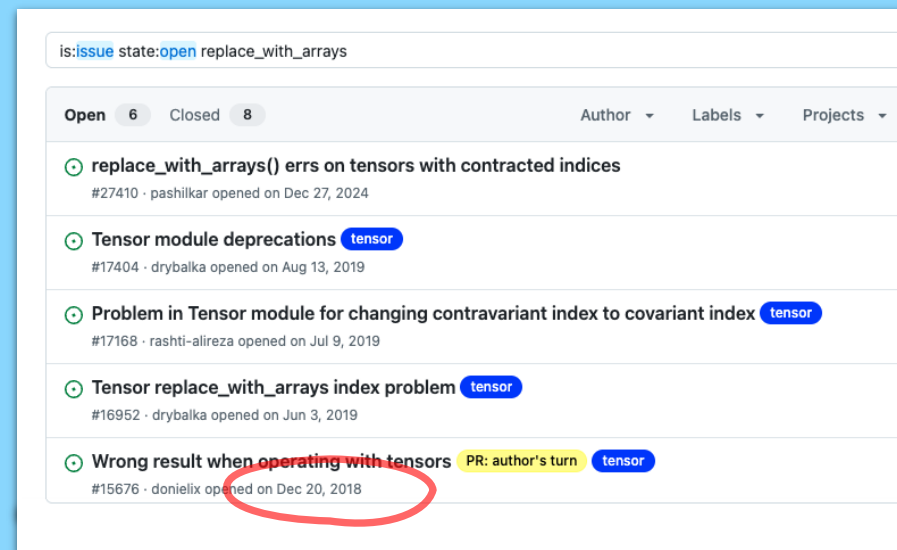
- a) **Momentum** – rank 1 vector, often denoted as $p(\mu) = (E, p_x, p_y, p_z)$
- b) **Spinor** – rank 1, denoted as U or V depends on momentum and spin
- c) **Metric** – rank 2 matrix, denoted as $g(\mu, \nu)$
- d) **Gamma matrixes** – rank 3, denoted as $\gamma(\mu) = (\gamma_0, \gamma_1, \gamma_2, \gamma_3)$

Fixed with one line of code

```
sympy/tensor/tensor.py

@@ -2210,7 +2210,7 @@ def _contract_and_permute_with_metric(metric, array, pos, dim):
2210 2210
2211 2211     array = tensorcontraction(tensorproduct(metric, array), (1, 2+pos))
2212 2212     permu = list(range(dim))
2213 -     permu[0], permu[pos] = permu[pos], permu[0]
2213 +     permu.insert(pos, permu.pop(0))
2214 2214     return permutedims(array, permu)
2215 2215
2216 2216     @staticmethod
```

This long-standing bug occurs when a metric is summed with tensors of rank higher than 2.



Fixed by Pull Request 28240

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